Adaptive versus Fixed Thresholds for Memory Filtering in MA-SAM2-Based Video Object Segmentation

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1. Problem and Motivation

Video object segmentation means identifying the pixels of a selected object in each frame of a video. For example, after a user selects a dog in the first frame, the model should keep finding the same dog as it moves, changes its appearance, or becomes partly hidden. Unlike a bounding box, a segmentation mask needs to follow the visible outline of the object.

SAM 2 uses information from earlier frames and predicted masks as memory to help process later frames (Ravi et al., ICLR 2025). MA-SAM2 improves memory handling for surgical videos. Its Context-Aware Memory (CAM) uses preset confidence and predicted-IoU thresholds to select reliable results (Yin et al., MICCAI 2025, Section 2.2). Predicted IoU is the model's estimate of mask quality and does not require the ground-truth test mask.

However, it is not clear whether the same fixed CAM thresholds are suitable for general videos with different objects and changing conditions. A strict threshold may reject useful updates, while a loose threshold may allow incorrect predictions into memory. My research question is: can an adaptive CAM predicted-IoU threshold improve segmentation compared with a tuned fixed threshold, using the same model and memory capacity?

To investigate this question, I will first study how sensitive segmentation performance is to the fixed threshold on DAVIS 2017. I will then propose an adaptive threshold based on recent model outputs and compare it with the fixed baseline. The aim is to understand when adaptation helps or fails, rather than assume that it will always perform better.

2. Data and Possible Obstacles

I plan to use the DAVIS 2017 semi-supervised TrainVal dataset at 480p resolution. It contains 90 video sequences: 60 for training and 30 for validation. The dataset provides RGB frames and pixel-level object masks. Unlike the surgical dataset, it contains general video scenes with different objects. The task is to follow each selected object instance across the video, rather than classify instrument categories.

I will select 10 sequences from the training split for development and use the 30 validation sequences for evaluation. The development sequences will be selected before comparing the methods. They will be used to choose the fixed threshold and the adaptive settings. The same first-frame object masks will be given to all methods, with no further user corrections during evaluation. The remaining ground-truth masks will only be used to measure performance. Frames from the same video will stay together, and their original order will be preserved.

The input and output format is summarised below.

Input	An ordered sequence of RGB frames and the target object masks in the first frame.
Label	A pixel-level mask identifying the individual objects and the background in each frame.
Output	A predicted mask for each target object in the following frames.

Possible obstacles include similar-looking objects, occlusion, fast movement and limited GPU memory. MA-SAM2 was designed for surgical videos, so I will first check whether its implementation and memory filtering work correctly on general videos.



Figure: An example RGB frame and its ground-truth object mask from DAVIS 2017.

3. Methods and Implementation

I will start from the official SAM 2 code and pretrained weights and the released MA-SAM2 predictor. I will first find compatible versions and run a short video. This project will change inference-time memory filtering; it will not train the backbone. I will check the implemented CAM gate against the paper before making changes.

Only the CAM predicted-IoU threshold will be changed. The confidence gate, occlusion-handling rules, prompts, preprocessing and memory capacity will stay the same. The comparison will include original SAM 2 as a reference, released MA-SAM2 settings, a development-tuned fixed threshold, and my adaptive threshold.

For the adaptive rule, I will keep a rolling history of up to 30 previous frame-level predicted-IoU scores, collected before admission filtering. Once ten previous scores are available, the threshold will be calculated as the recent median minus a small multiple of the median absolute deviation, which measures score spread. The threshold will be clipped to a predefined range. Before ten scores are available, I will use the tuned fixed threshold. The rule will only use past model outputs, not future frames or ground-truth test masks. I will check how the frame-level score is calculated in the released implementation.

As an initial plan, fixed thresholds will be tested at 0.60, 0.70, 0.80, 0.90 and 0.95, plus the released setting, if the implemented score range supports this grid. The adaptive rule will start with bounds of 0.60 and 0.95, and test three multipliers: 0.5, 1.0 and 1.5. All choices will use development videos only and be locked before evaluation. I will report the search cost. Related work must still be checked before claiming that this rule is new.

4. Evaluation and Expected Results

Each method will receive the same first-frame ground-truth mask for each target object. No further user corrections will be provided during evaluation, and the first frame will not be scored. Model size, hardware, preprocessing and memory capacity will be kept the same across methods.

Evaluation	Report
Segmentation quality	Primary metric: mask IoU. Average over frames within each target, then targets within each video, then equally over videos. Exclude and count empty-empty cases. Also report false-positive area when the target is absent.
Comparison and cost	I will report paired per-video IoU differences, time per frame and peak GPU memory on the same machine. An exploratory 95% paired bootstrap confidence interval will be calculated by resampling the 30 evaluation videos, rather than individual frames.
Plots and examples	Fixed-threshold sensitivity plot, per-video comparison, adaptive-threshold and admission-rate curves, and side-by-side mask overlays showing both good and bad cases.

For a small exploratory analysis, I will mark occlusion/reappearance intervals before comparing methods and inspect the five annotated frames after reappearance. I will state the number of cases and distinguish occlusion from leaving the camera view where possible. I will also inspect which frames enter memory and whether they contain poor masks.

My hypothesis is that bounded adaptive filtering may help when prediction quality changes over time. However, it may accept more errors after a long failure, or offer no benefit over a tuned fixed threshold. Negative or mixed results will still help explain when adaptation is useful.

After completing the core experiments, I plan to build a small desktop viewer using C# and Avalonia to compare saved predictions and inspect selected memories. Python will run the inference. The viewer supports the study but is not the main research contribution. Priority will be baseline setup, threshold analysis, adaptive experiments and then the viewer.

Related Conference Paper and Github Repository:

SAM 2:

Paper: https://arxiv.org/pdf/2408.00714

GitHub Repo: https://github.com/facebookresearch/sam2

MA-SAM2:

Paper: https://papers.miccai.org/miccai-2025/paper/2634_paper.pdf

GitHub Repo: https://github.com/Fawke108/MA-SAM2

Datasets:

Official Page: https://davischallenge.org/davis2017/code.html

Kaggle: https://www.kaggle.com/datasets/lennelenne/davis2017